

Day 6.6 - Shared Held-out Evaluation

PretrainLab-X | final evidence report | 2026-09-13

Outcome. The existing Day 5 and Day 6 336M checkpoints were evaluated on one shared held-out stream. No model weights, optimizer state, or training history were modified. The Day 6 fresh-data checkpoint has substantially lower held-out NLL and perplexity under the matched protocol.

1. Protocol and comparability

Control	Fixed value
Model	336,118,784 parameters; dim 1024; 24 layers; 16 Q heads; 4 KV heads; RMSNorm, RoPE, SwiGLU; context 512
Tokenizer	TinyLlama/TinyLlama-1.1B-Chat-v1.0; vocabulary 32,000; EOS id 2
Evaluator	Same script; model.eval(); torch.inference_mode(); BF16 autocast; no backward pass or optimizer
Loss	Token-weighted causal cross-entropy; one next-token target is omitted at each document boundary
Hardware	NVIDIA RTX PRO 6000 Blackwell Server Edition (CUDA)

Both checkpoints are at step 15,259 and have the same parameter count and architecture. The intentional training-history difference is Day 5 repeated API-sourced data versus Day 6 fresh-shard data.

2. Shared held-out stream

Source: FineWeb-Edu sample-10BT, shard 013_00000.parquet. The builder inspected 8,681 rows and kept 8,666 documents, producing 10,000,094 tokens. The requested evaluation budget was 10,000,000 tokens; 9,992,428 were scoreable after document-boundary masking. Candidate-level exact text deduplication removed zero rows.

Token-stream SHA-256: 3fd022ee8933fc828a8876bf044ef7b40f96bf8cb0f98509a21df9214a6c750

Overlap audit: a 64-token SHA-256 window screen at stride 16 was run against the reconstructable Day 5 9.9M-token stream; 15 candidate documents were rejected. The candidate shard is different from Day 6 training shard 000_00000.parquet. Day 5 was obtained through the dataset-server API without retained shard/URL identifiers, so exact Day 5 document/URL reconstruction and semantic zero overlap are not proven.

3. Matched evaluation results

Checkpoint	Tokens	Docs	Weighted NLL	Perplexity	Time (s)	Tok/s
Day 5 repeated API	9,992,428	8,666	8.4242114093	4,556.0505025110	62.649	159,497.94
Day 6 fresh shard	9,992,428	8,666	3.0394002921	20.8927099547	62.418	160,089.71

Raw NLL sums: Day 5 = 84,178,325.96462083; Day 6 = 30,370,988.58282305. Peak allocated GPU memory was 2,614.41 MiB for each run.

4. Paired document bootstrap

The 8,666 shared document IDs were resampled 10,000 times with seed 1337, preserving document token counts. The Day 5 minus Day 6 weighted-NLL difference is **5.3848111722**; percentile 95% interval [**5.3670995529**, **5.4028614536**]. Perplexity difference is 4,535.1577952563; Day 6 perplexity is 99.5414% lower relative to Day 5.

This is controlled held-out evidence, not a causal claim that data freshness alone explains the gap: the checkpoints also have different training histories.

5. Reproducibility artifacts

Repository implementation:

evaluation/build_heldout.py - shared stream builder and overlap audit
evaluation/shared_heldout_eval.py - identical checkpoint evaluator
evaluation/compare_bootstrap.py - paired document bootstrap
evaluation/build_audits.py - checkpoint, comparability, and overlap manifests
report/day6_6_summary.json - compact machine-readable result summary

Remote evidence retained on the AutoDL data disk: heldout_manifest.json, heldout_tokens.int32, heldout_doc_index.jsonl, day5_formal_10m.json, day6_formal_10m.json, paired_bootstrap_10k.json, checkpoint_manifest.json, comparability_audit.json, and overlap_audit.json.

Checkpoint SHA-256 (Day 5): 2c49925b61e46c533026c201dba02bfe845071e7c453964fcc33f508a24c36f3

Checkpoint SHA-256 (Day 6): 3d1250567b0d3f311584b15a3acd173539cdd52b0baebfd7b290e1f7fdf21aca

6. Cost and cleanup boundary

The GPU was used only for held-out inference (two formal passes plus a 100K-token smoke check); no additional training was launched. The 516 MB raw parquet candidate was deleted after its processed token stream and manifest were verified. The AutoDL instance should be released immediately after this report is checked; final billing is determined by the platform hourly meter.

Conclusion: Day 6.6 is complete. The result is a reproducible, matched held-out comparison with explicit overlap and causality limitations.